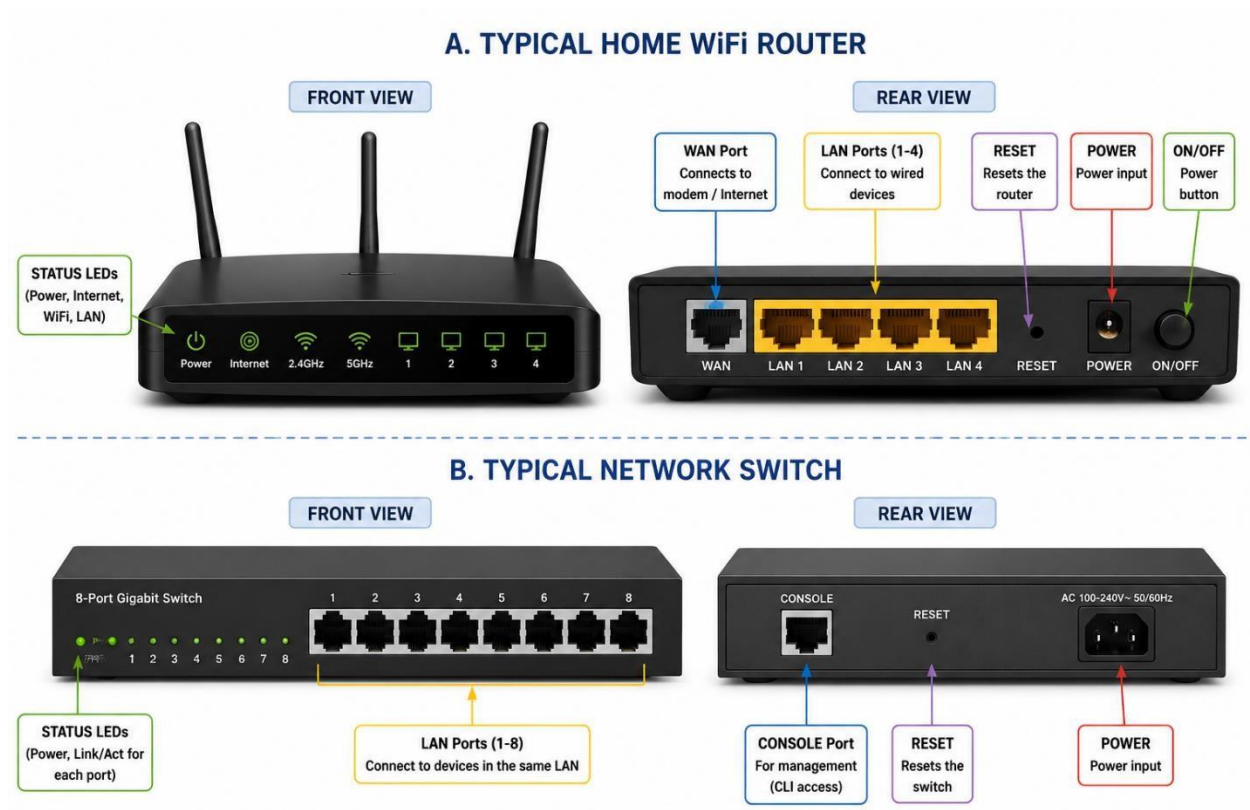


1.Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



2.List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Feature	Router	Switch
Main Function	Connects different networks (such as a home network to the Internet).	Connects devices within the same Local Area Network (LAN).
How It Handles Traffic	Uses IP addresses to route data only to the correct device on the local network.	Uses MAC addresses to forward data packets between different networks.
Network Type	Connects LAN to WAN (Internet) and manages communication to the Internet by itself.	Connects multiple devices within a single LAN and does not route traffic between networks.

3.Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



Router

Identify these parts:

- Power Port – Located on the back of the router. Used to connect the power adapter.
- Ethernet Ports (LAN 1 & LAN 2) – Located on the back. Used to connect devices such as a desktop PC, smart TV, or gaming console.
- LED Indicators – Located on the front. Common LEDs include Power, Internet, WiFi (2.4 GHz/5 GHz), and LAN status. They show whether the router and connections are working properly.

Network Switch

Identify these parts:

- Power Port – Located on the back (or side on some models). Used to supply power to the switch.
- Ethernet Ports (Port 1 & Port 2) – Located on the front. Used to connect computers, printers, or other network devices.
- LED Indicators – Located on the front, above or beside each Ethernet port. These LEDs indicate Power, Link, and Activity (blinking means data is being transmitted).

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

For a small gaming café, I would connect the router first to the ISP line because it provides internet to the network. Then, I would connect an 8-port switch to the router and connect all 8 gaming PCs to the switch using CAT6 Ethernet cables.

This setup gives a fast, stable, and reliable wired connection, which is best for LAN gaming and IPL streaming.

Network Connection:

ISP Line → Router → 8-Port Switch → 8 Gaming PCs